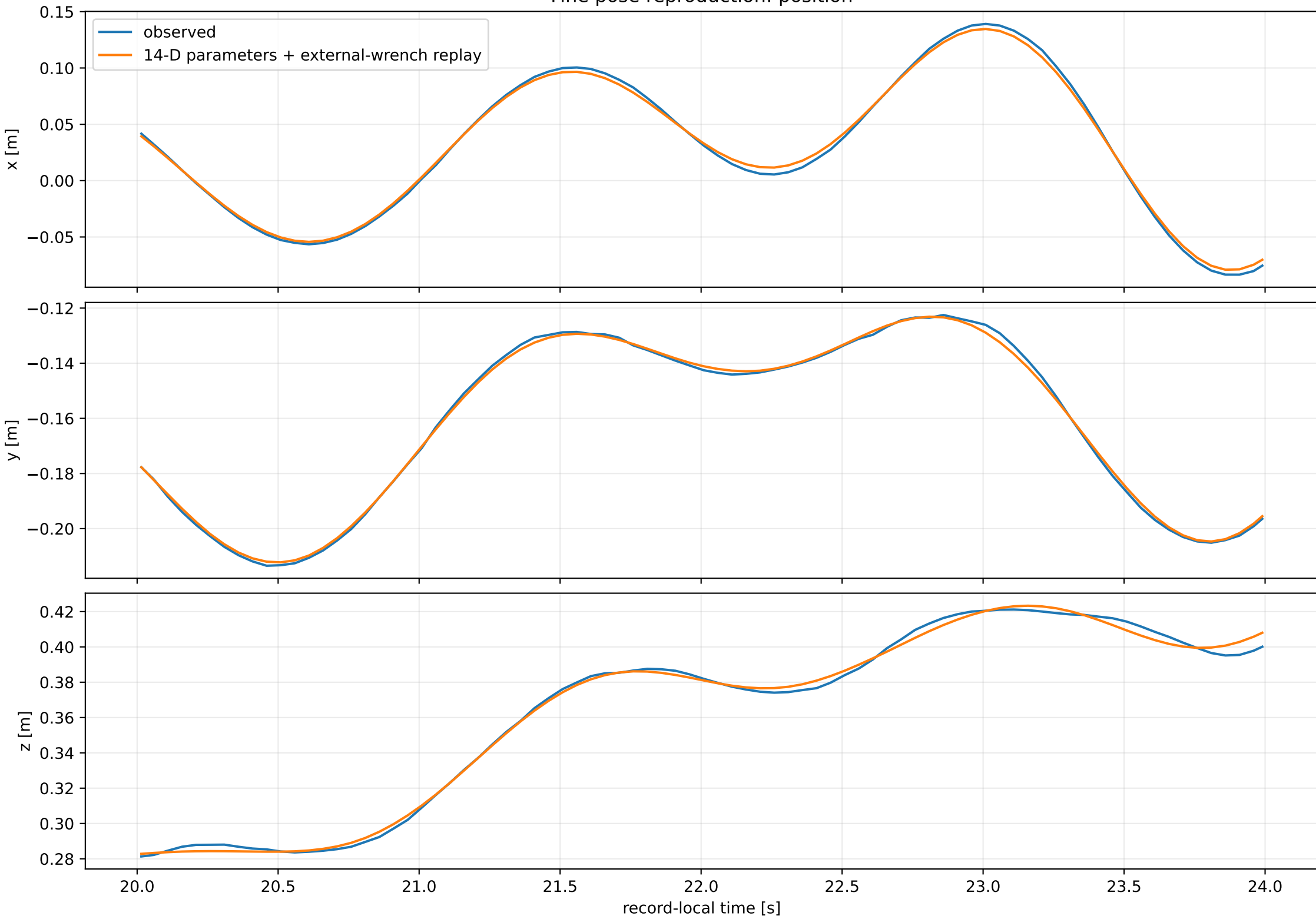
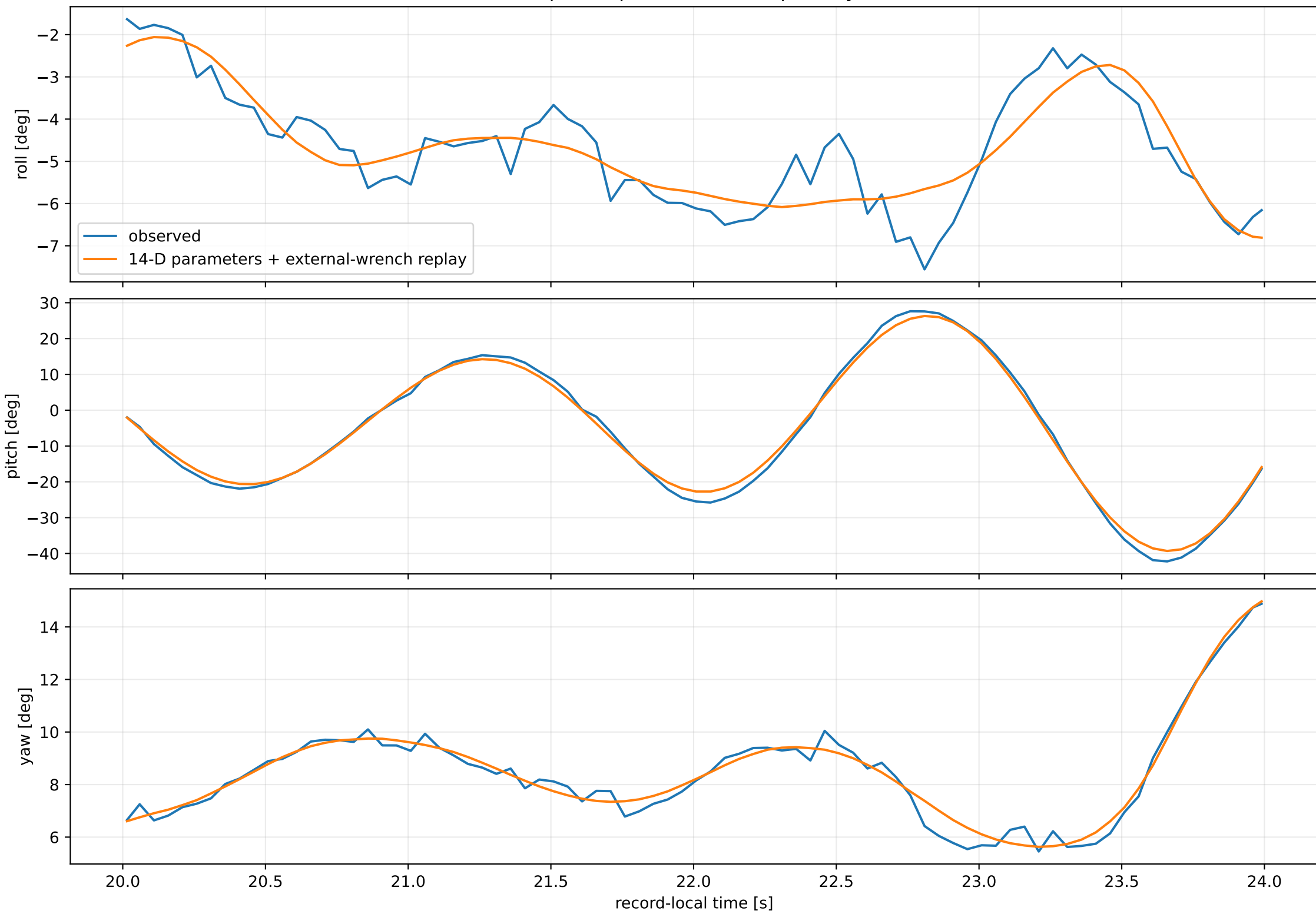


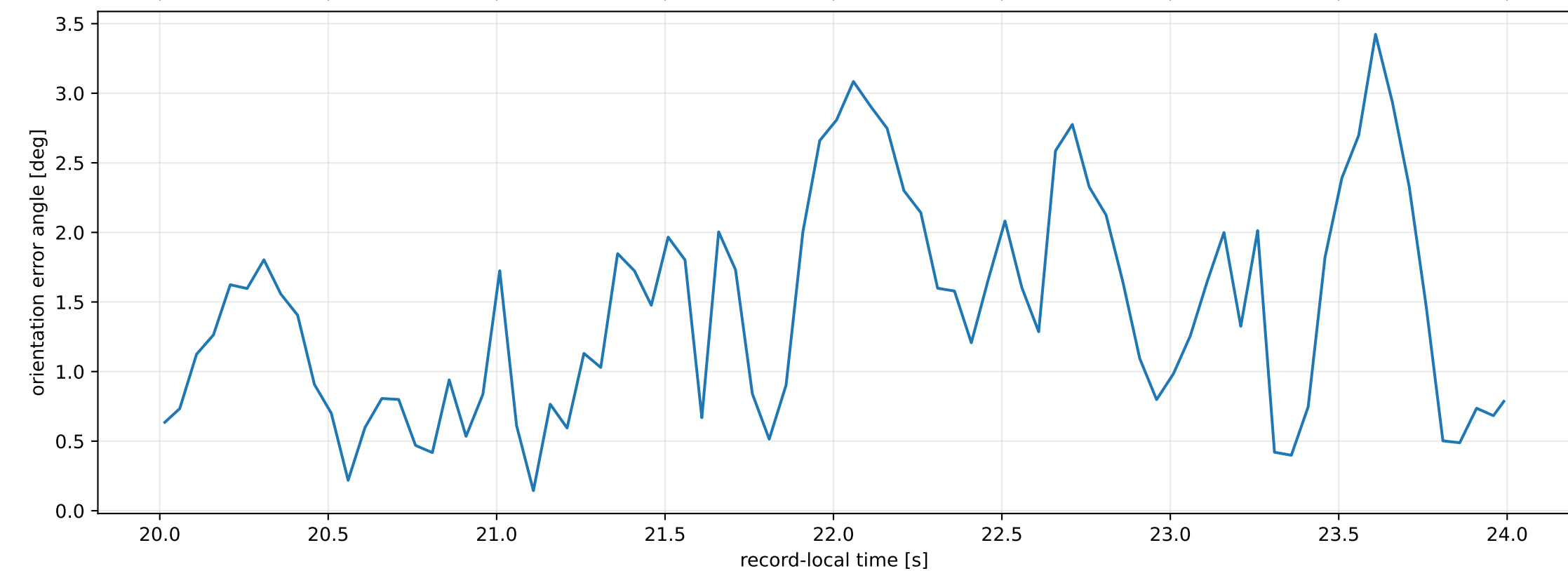
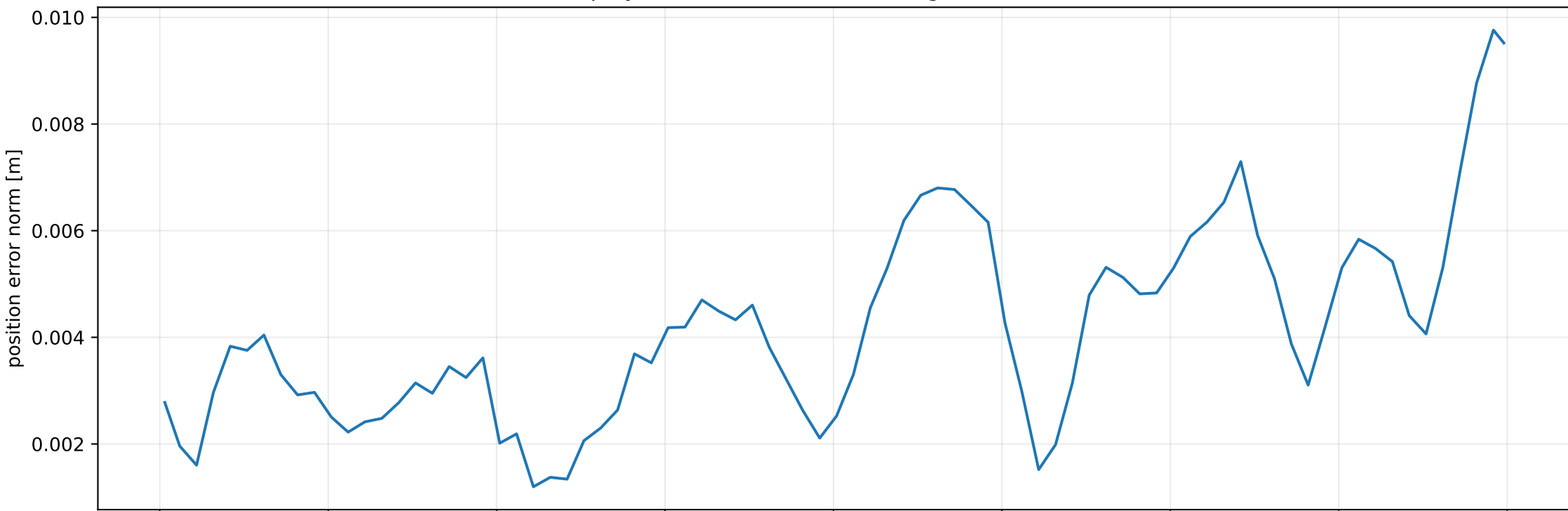
Fine pose reproduction: position



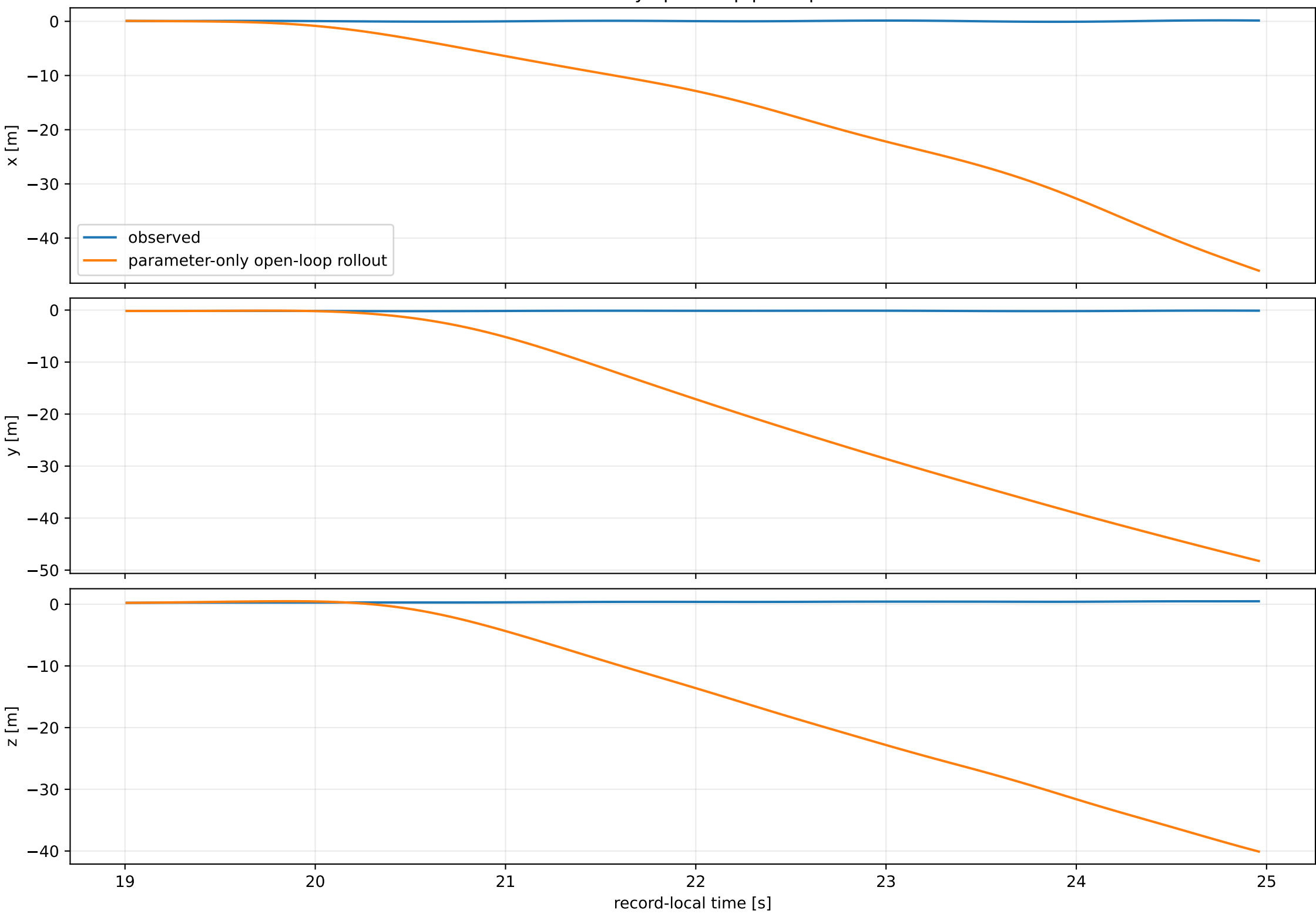
Fine pose reproduction: roll / pitch / yaw



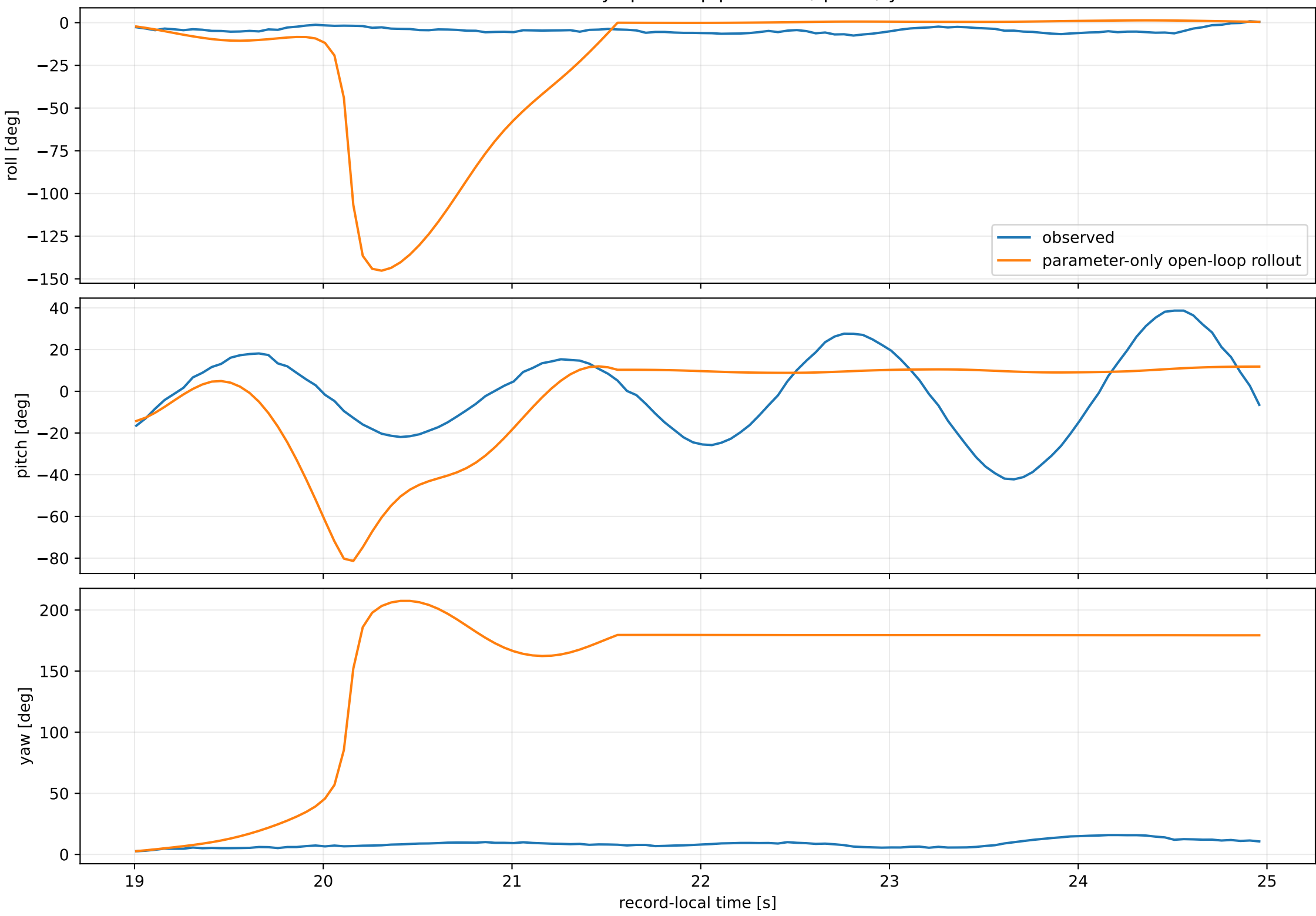
Fine replay error after the 14-D change — lower is better



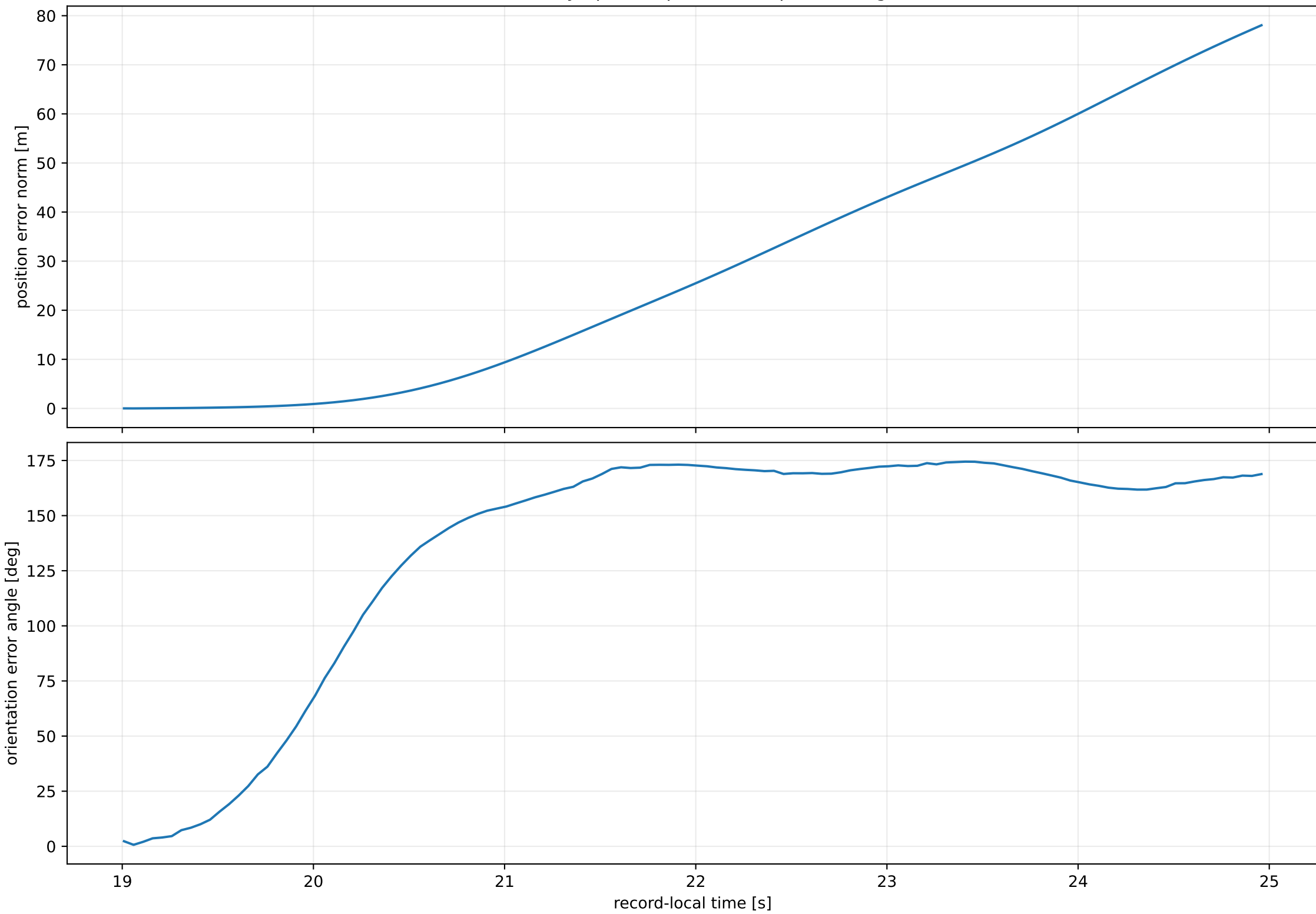
Parameter-only open-loop pose: position



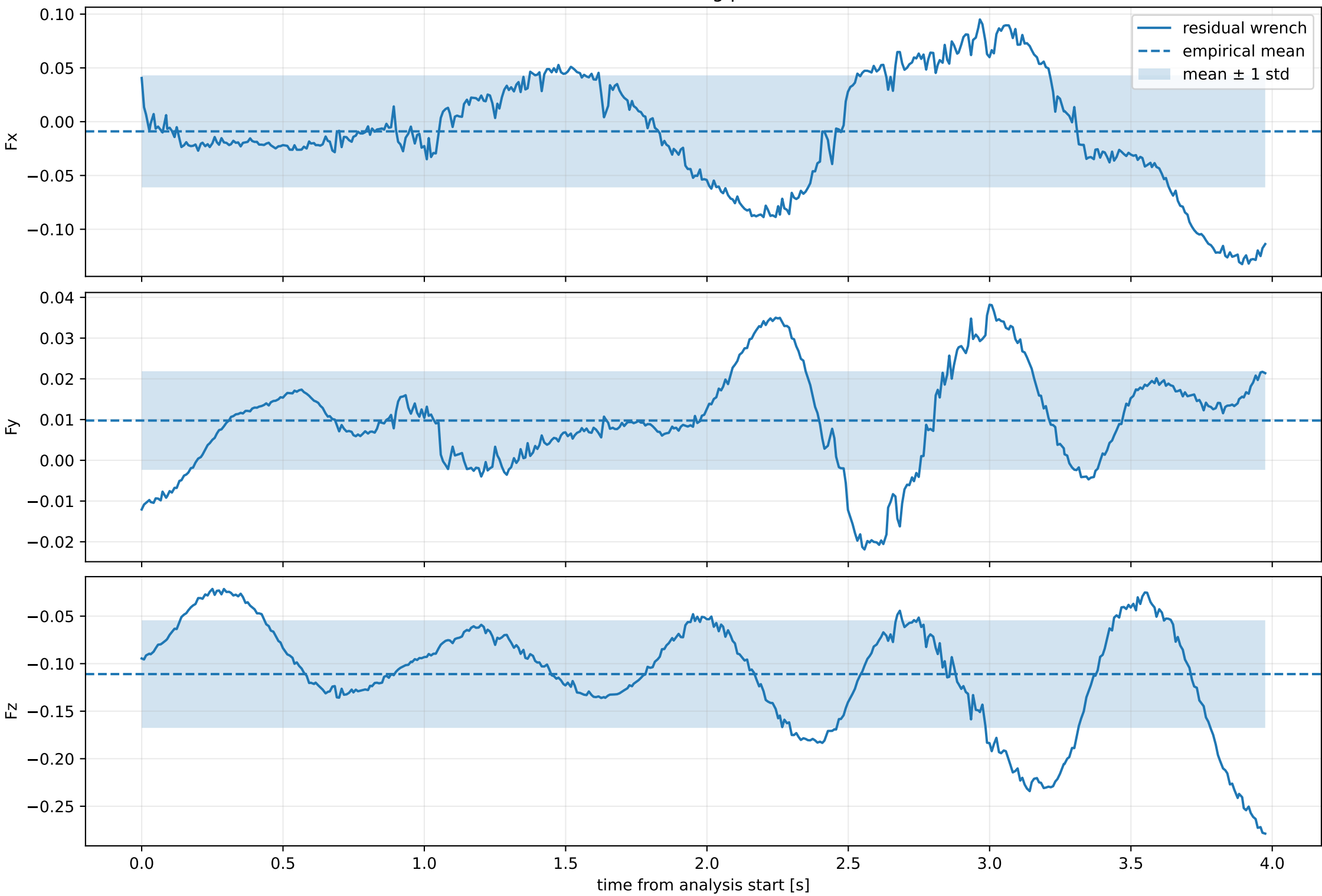
Parameter-only open-loop pose: roll / pitch / yaw



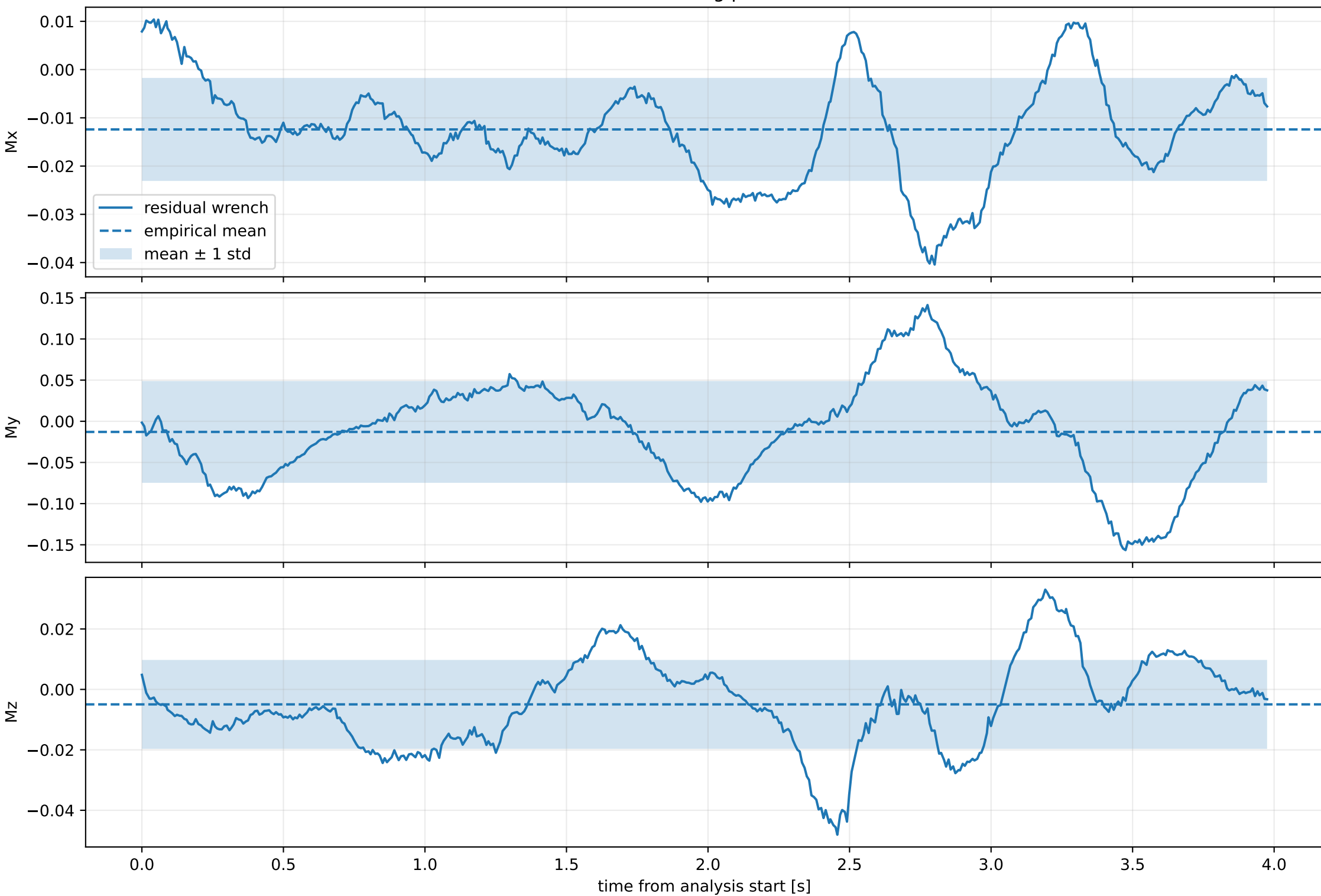
Parameter-only open-loop rollout (separate diagnostic)



Dimensionless residual force: empirical mean  $\pm$  1 std  
Wide band = disturbance varies more, so matching parameter directions receive weaker evidence

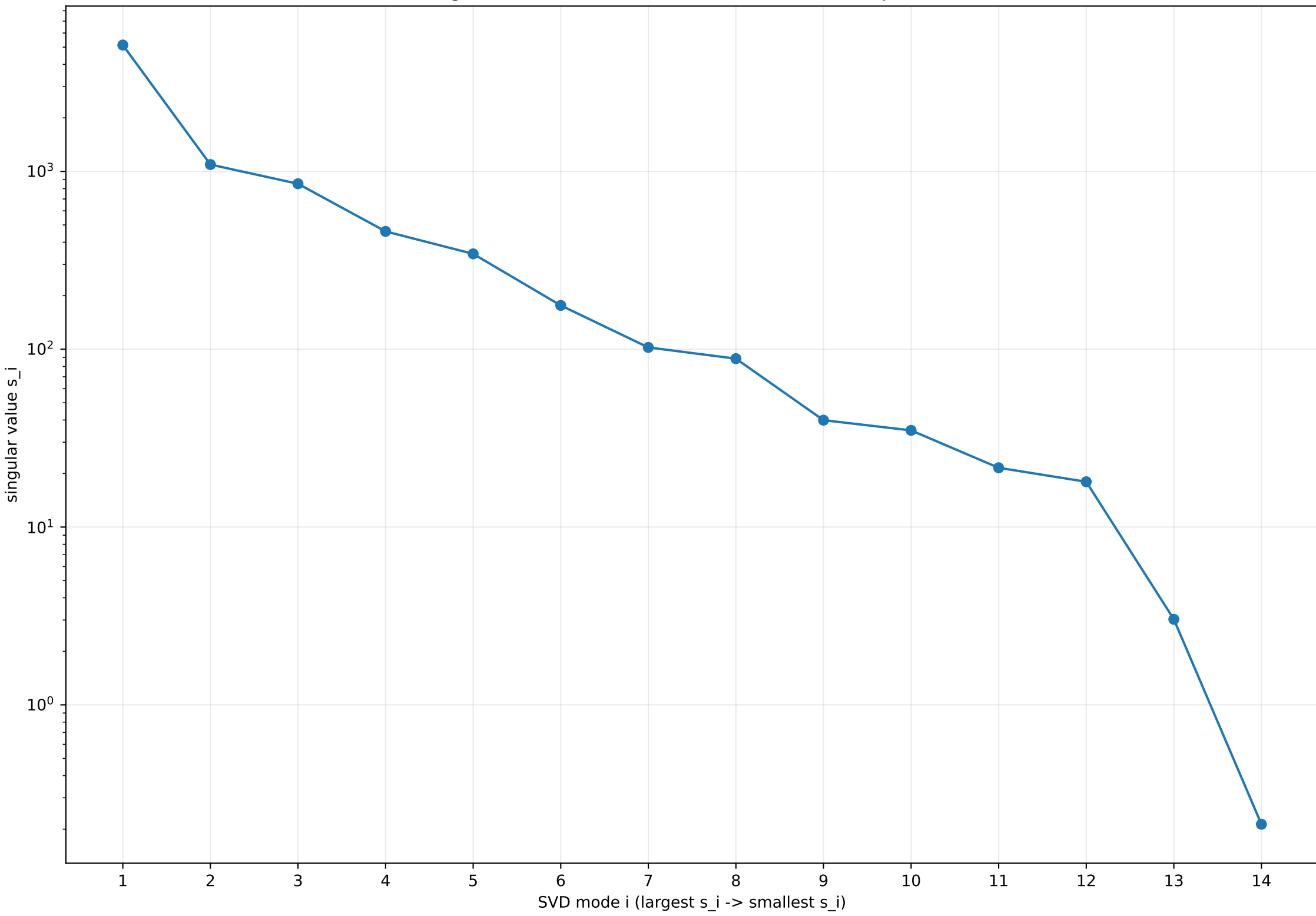


Dimensionless residual torque: empirical mean  $\pm 1$  std  
Wide band = disturbance varies more, so matching parameter directions receive weaker evidence

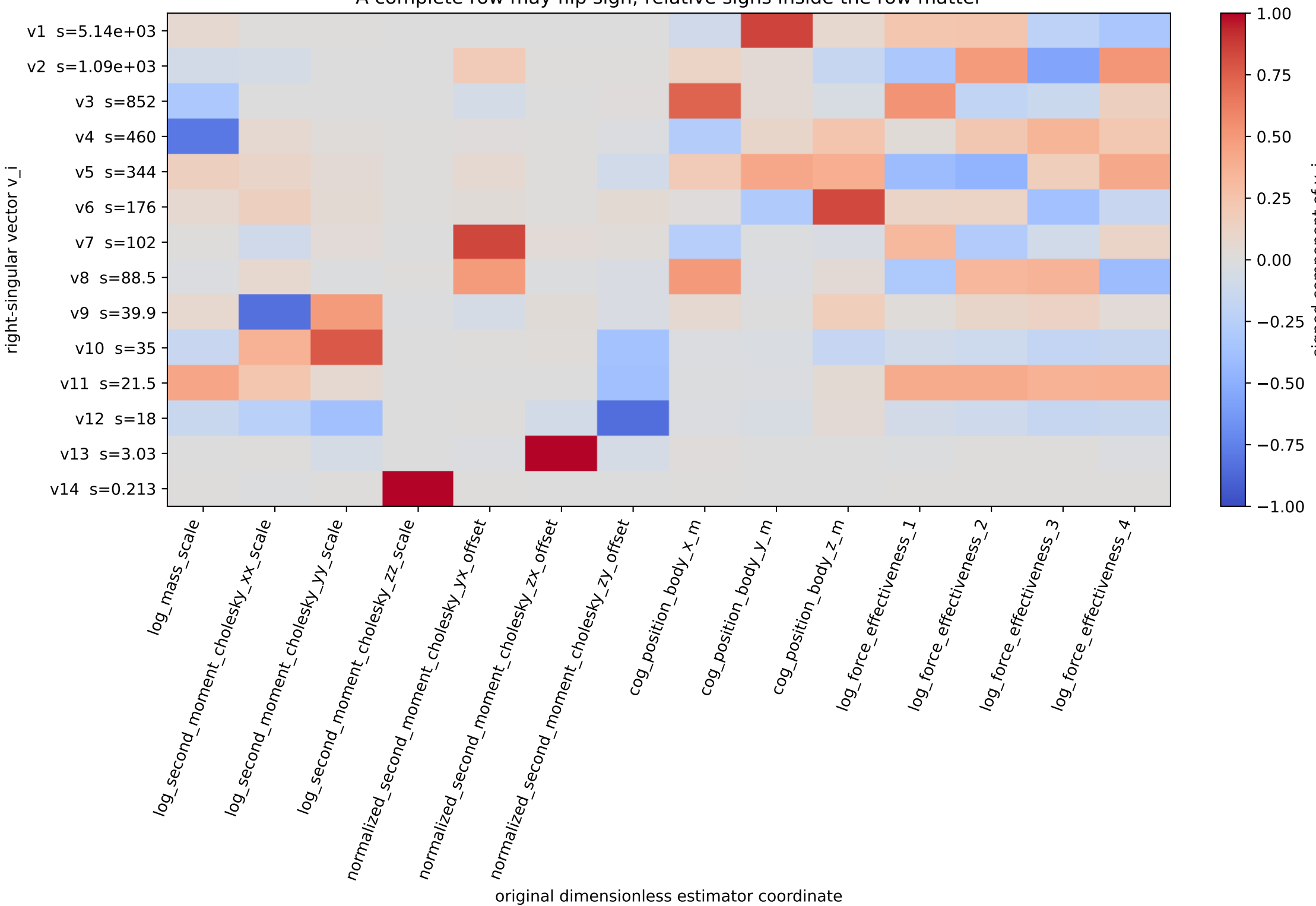




Singular values  $S$  of the whitened dimensionless Jacobian



Signed right-singular vectors  $V^T$   
A complete row may flip sign; relative signs inside the row matter



Decoded physical parameters at the selected estimate

mass [kg]

2.36568466864

inertia J [kg m<sup>2</sup>]

[ 0.065013604 -1.7771145e-07 2.4009695e-05]  
[-1.7771145e-07 0.065299532 1.097643e-05]  
[ 2.4009695e-05 1.097643e-05 0.12899452]

principal moments [kg m<sup>2</sup>]

[ 0.065013594, 0.06529953, 0.12899453]

CoG position in body frame [m]

[-0.0028362247, 0.00085194641, 0.010886318]

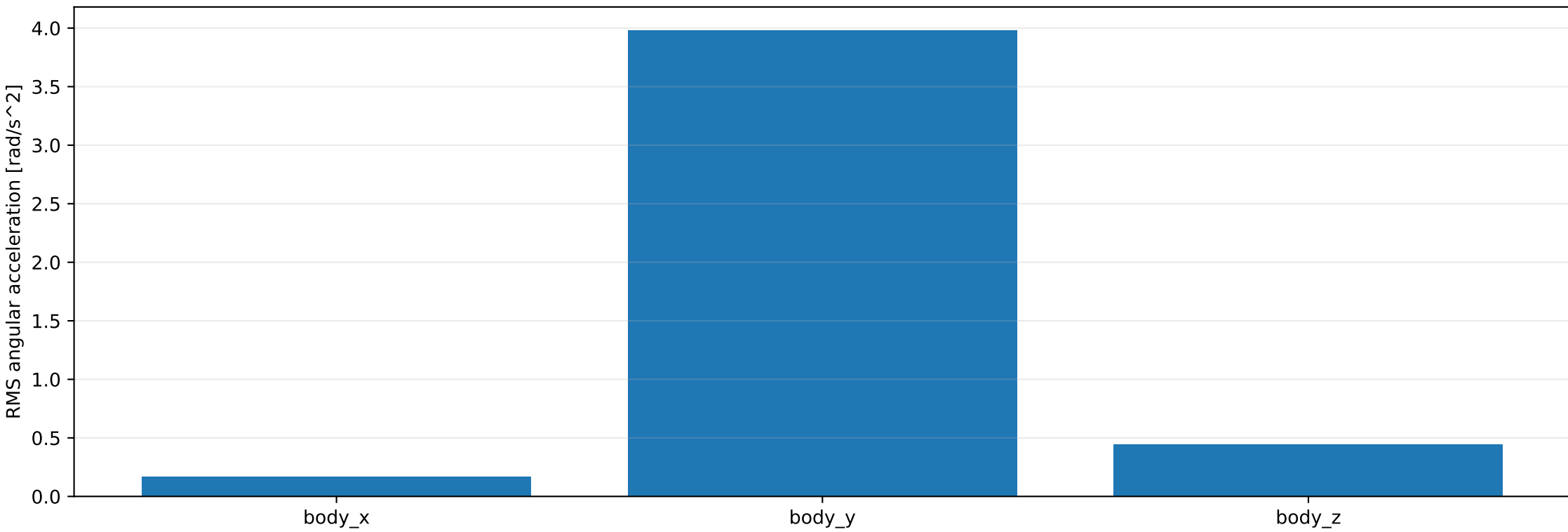
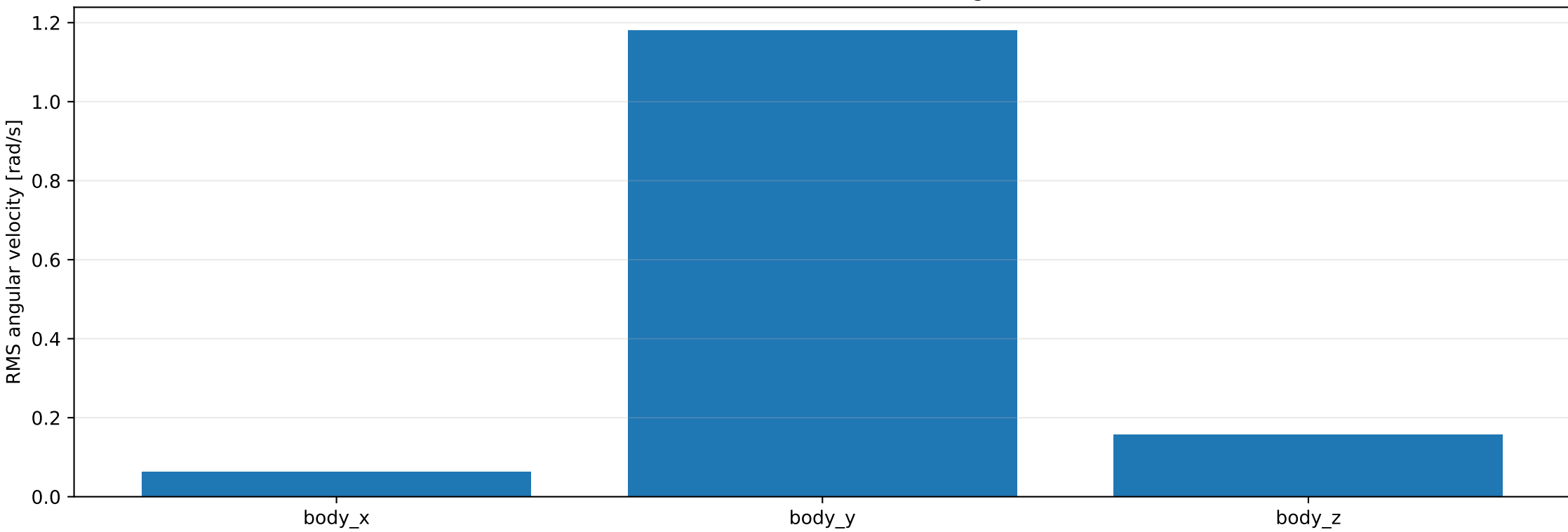
rotor force effectiveness

[ 0.96281281, 0.97499493, 0.96194124, 0.95161385]

selected delay [s]

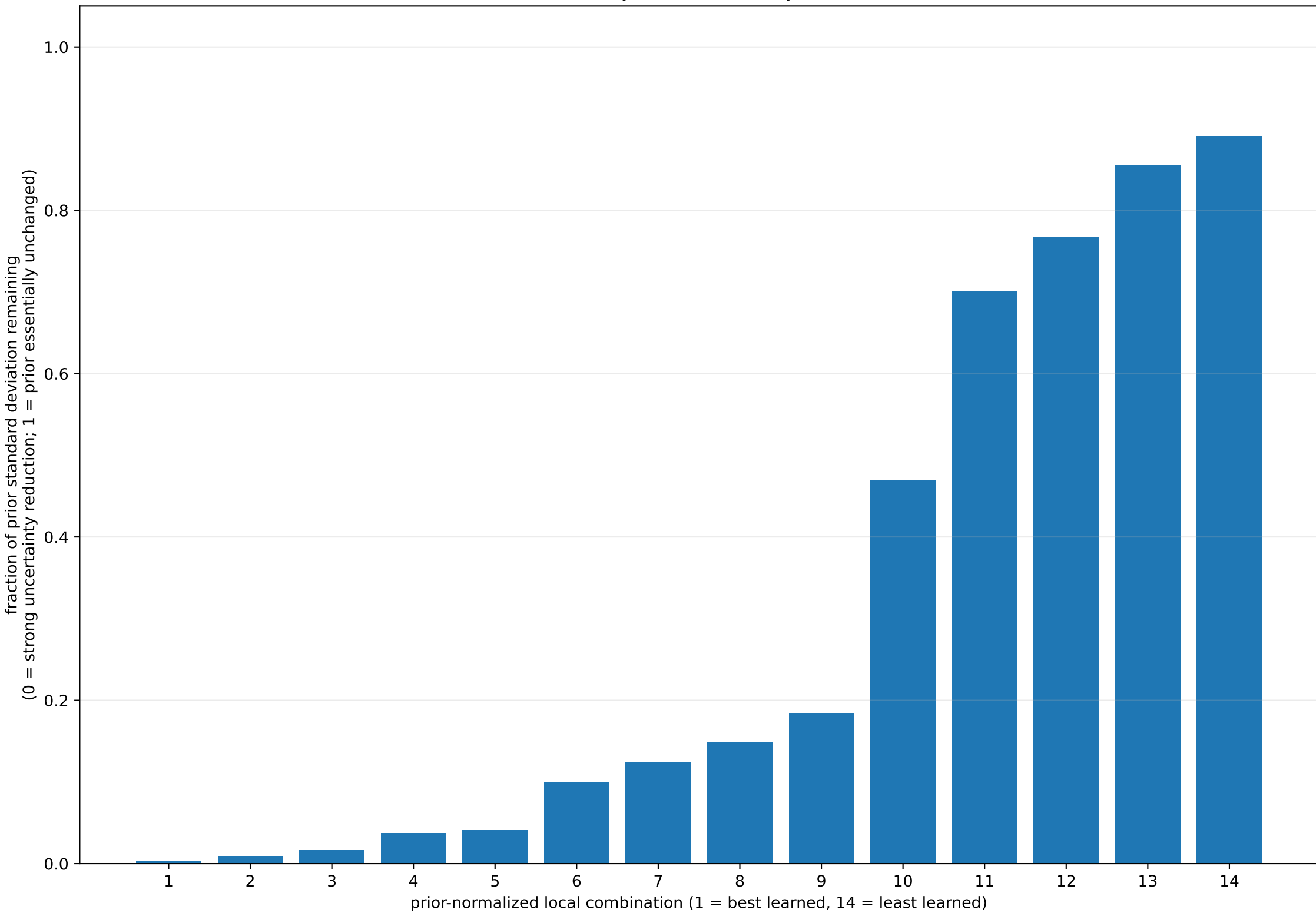
0.2

Axis-wise rotational excitation diagnostic



Low body-z values can support a yaw-excitation hypothesis, but identifiability is decided by the full Jacobian.

How much uncertainty remains in every local combination?



Confidence accumulation as the rosbag becomes longer

